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School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

Date: 09-03-2026

Full-Stack Skill Learning Academy Marketplace With Secure Authentication

SDG: SDG 4 – Quality Education

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INTRODUCTION



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Online learning platforms have become an essential part of modern education. Students require flexible access to learning resources such as video lectures, notes, and quizzes from anywhere.

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Traditional classroom learning alone cannot always meet the needs of today's digital learners.

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Therefore, an efficient online learning system is necessary to support interactive and accessible

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education. SkillForge is an online learning management system designed to provide structured digital

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learning. It allows users to enroll in courses, watch video lessons, download notes, and track their

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learning progress. The system also includes quizzes and certificate generation to evaluate student

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performance. Overall, SkillForge aims to create an easy, organized, and effective learning environment

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for students.

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PROBLEM STATEMENT



1

Existing learning systems often lack a centralized platform for managing courses, lessons, and study materials efficiently. Many students face difficulty accessing organized learning resources in one place.

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Traditional systems may rely on manual processes for managing course content and tracking student progress. This leads to inefficiency in delivering educational materials and monitoring learning

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outcomes. Students also require flexible learning environments that allow them to access courses

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anytime and anywhere. Therefore, there is a need for a simple, reliable, and automated online learning

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management system. The proposed system aims to provide structured course management with video

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lessons, downloadable notes, quizzes, and progress tracking. This system improves accessibility,

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organization, and effectiveness of digital learning.

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OBJECTIVE(S)



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Objective 1:

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To design and develop an online learning management system that allows users to access courses, video lessons, notes, and quizzes in a structured way.

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Objective 2:

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To implement a system that tracks student learning progress and improves accessibility to digital learning resources.

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PROPOSED SYSTEM



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- The proposed system is an online learning platform that allows students to access courses, video lessons, notes, and quizzes in a structured way.

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- It provides a centralized system where users can easily enroll in courses and start learning from anywhere.

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- Admins can manage courses , upload lessons, videos and study materials through an admin panel.

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- Students can watch video lectures , download notes , and complete quizzes for better understanding.
- The system tracks student progress by marking lessons as completed.
- It reduces manual management of course materials and improves organization of learning content.
- The platform provides a user-friendly interface for easy navigation and learning.

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PROPOSED SYSTEM ARCHITECTURE

Input

User registration and login details.

Course selection and enrollment by students.

Admin inputs course details, lessons, videos, notes, and quizzes.

Student interactions like watching videos and marking lessons as completed.

Processing

The system authenticates users during login.

It processes course enrollment and lesson access.

Tracks user learning progress and completed lessons.

Manages course content uploaded by the admin.

Database

Stores user information, login credentials, and profiles.

Stores courses, lessons, videos, notes, and quiz details.

Maintains student progress and enrollment data.

Database used: **MySQL**.

Output

Displays available courses to users.

Shows video lessons, downloadable notes, and quizzes.

Displays progress tracking and completion status.

Provides certificates or completion confirmation.

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PROPOSED SYSTEM ARCHITECTURE



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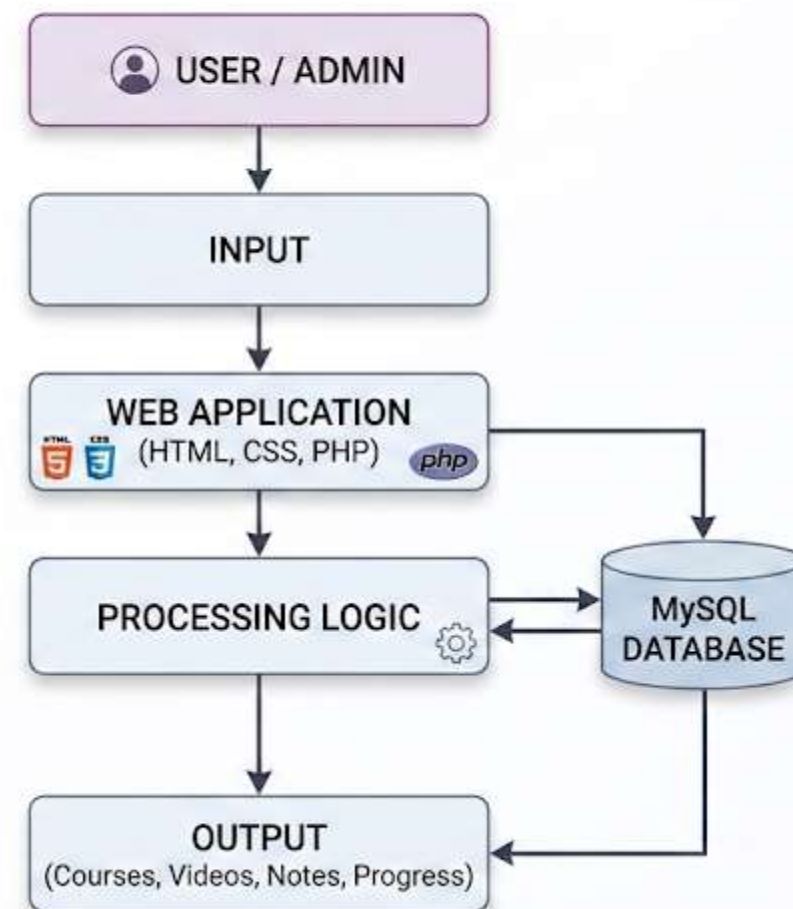
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BLOCK DIAGRAM: ONLINE LEARNING PLATFORM SYSTEM ARCHITECTURE



WORKFLOW



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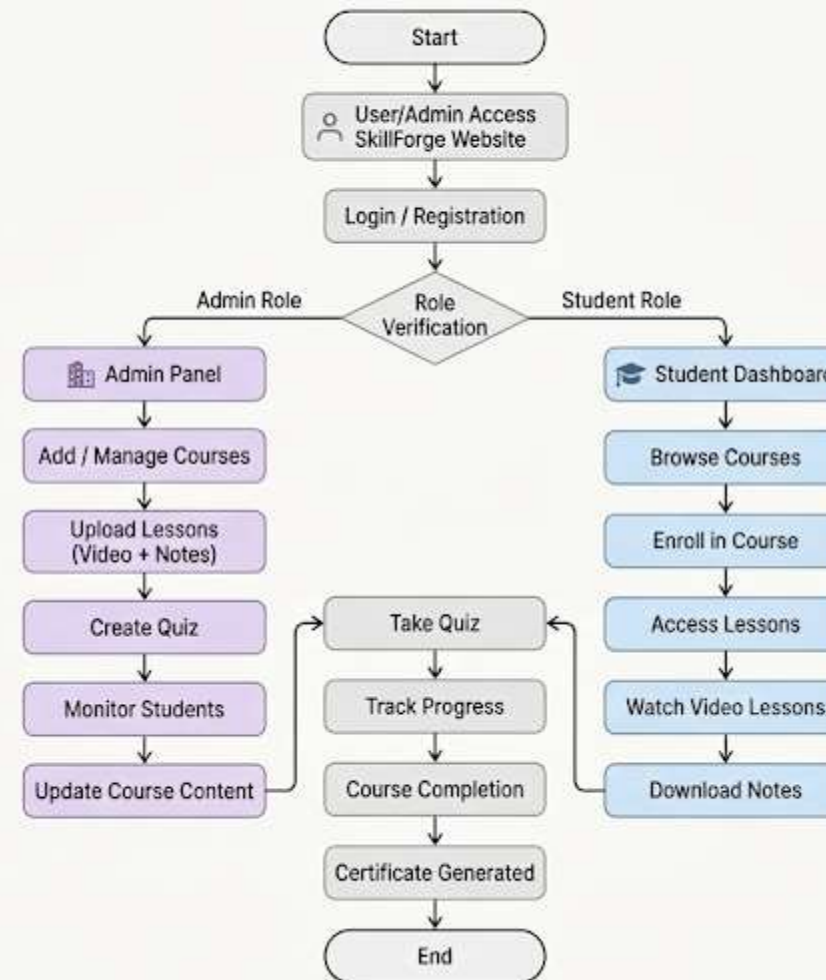
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TECHNOLOGIES USED



Frontend:

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HTML : Structure of the web pages

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CSS : Styling and layout design

JavaScript : Client-side interactivity and dynamic content

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Backend:

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PHP : Server-side scripting to handle application logic, user authentication, course management, and data processing.

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Database:

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MySQL– Used to store and manage system data

Tools:

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XAMPP – Local server environment for running Apache, PHP, and MySQL

Visual Studio Code – Code editor for development

phpMyAdmin – Database management tool for MySQL

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Web Browser (Chrome/Edge) – For testing and running the application

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MODULE DESCRIPTION



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Module 1: User Management

Handles user registration, login, and authentication for both students and administrators..

Module 2 Course Management

Allows the administrator to create, update, and manage courses.

Module 3: Learning & Enrollment System

Enables students to browse available courses and enroll in them.

Module 4: Quiz & Progress Tracking

Provides quizzes for evaluating student understanding after lessons.

Module 5: Certificate Generation

Generates certificates automatically when a student completes all lessons and assessments in a course.

IMPLEMENTATION SCREENSHOTS



Login page

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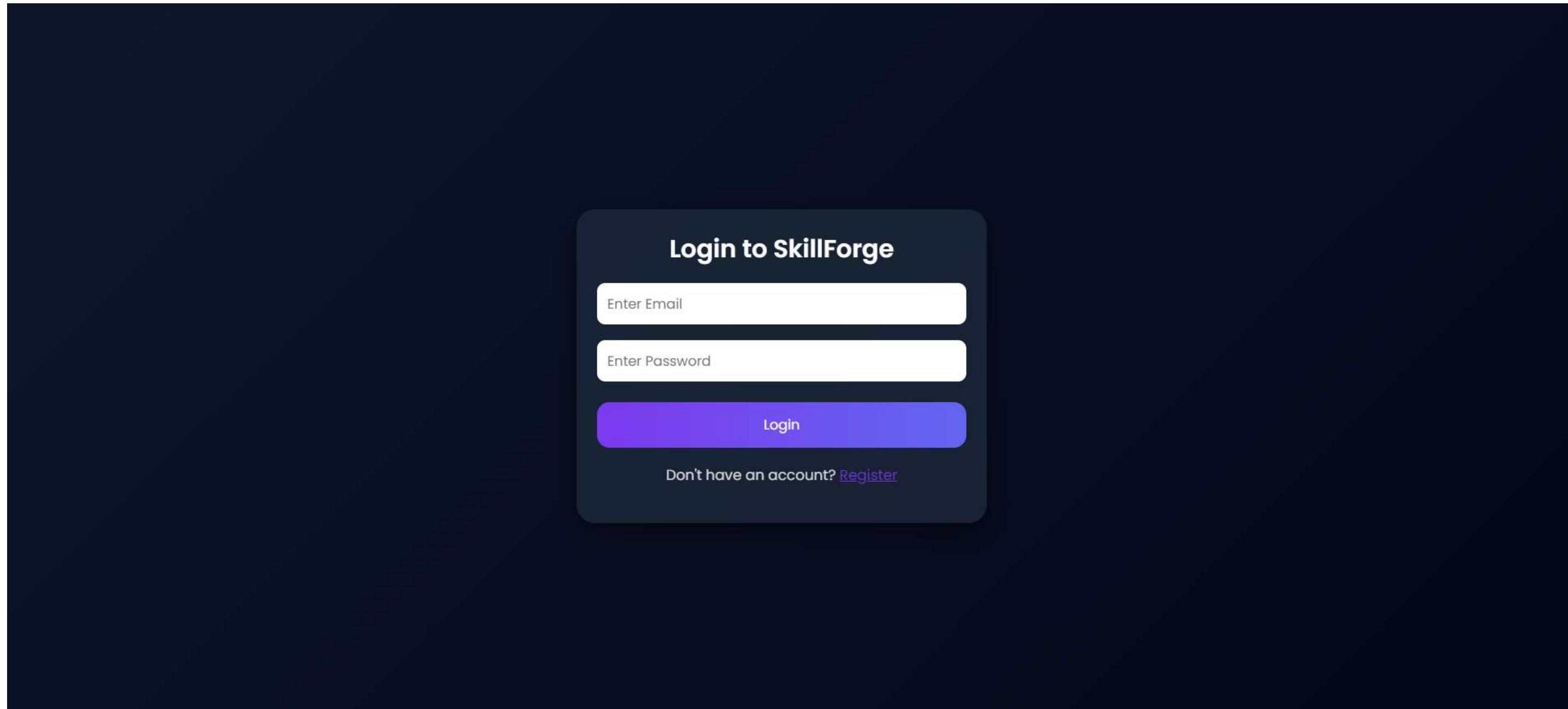
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A screenshot of a login page for 'SkillForge'. The page has a dark blue background. In the center, there is a dark blue rounded rectangle containing the login form. The form has the title 'Login to SkillForge' in white. Below the title are two white input fields: 'Enter Email' and 'Enter Password'. Below these fields is a purple 'Login' button. At the bottom of the form, there is a link that says 'Don't have an account? Register'.

Login to SkillForge

Enter Email

Enter Password

Login

Don't have an account? [Register](#)

IMPLEMENTATION SCREENSHOTS



Admin Home Page:

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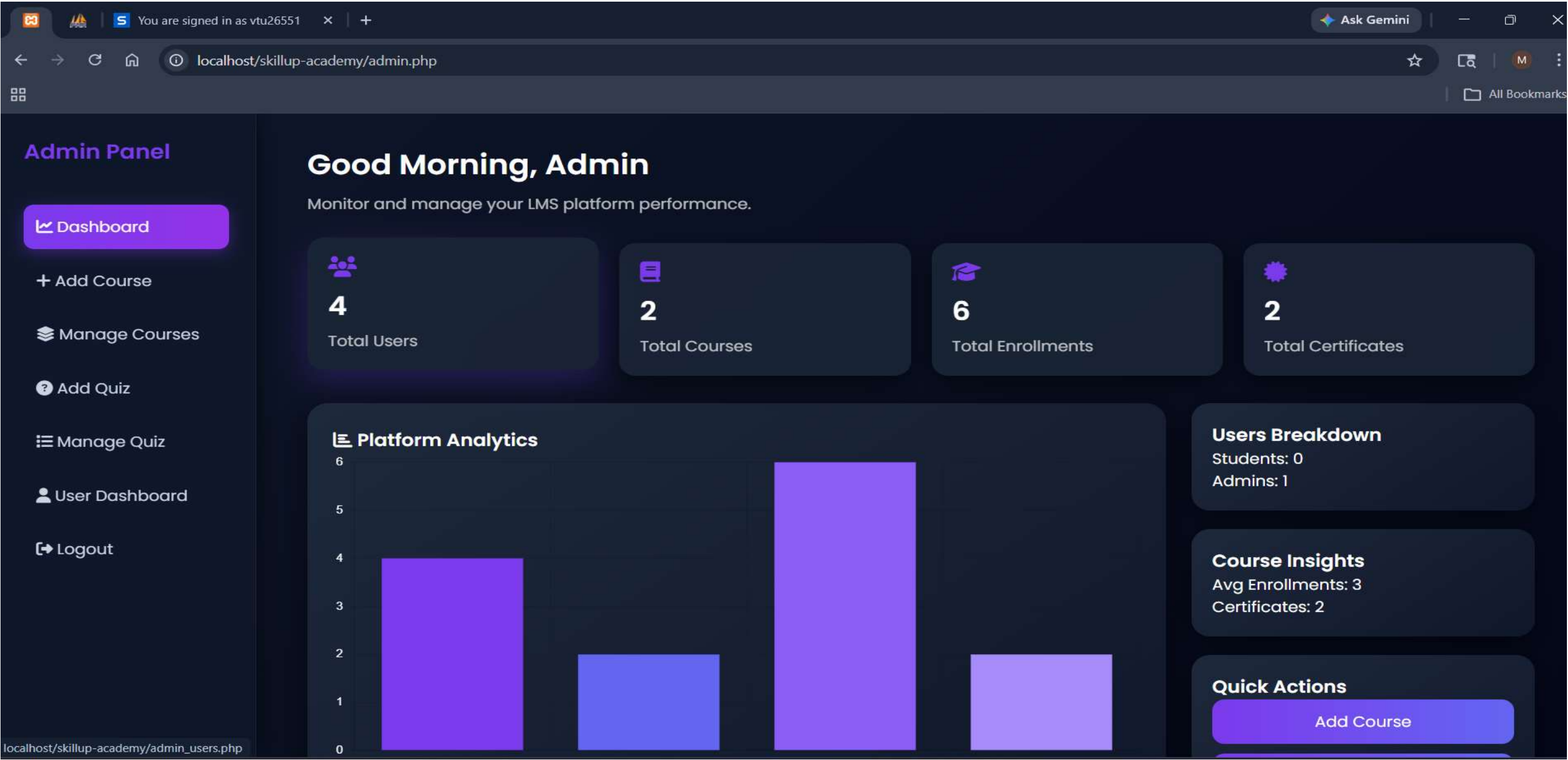
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IMPLEMENTATION SCREENSHOTS



User home page:

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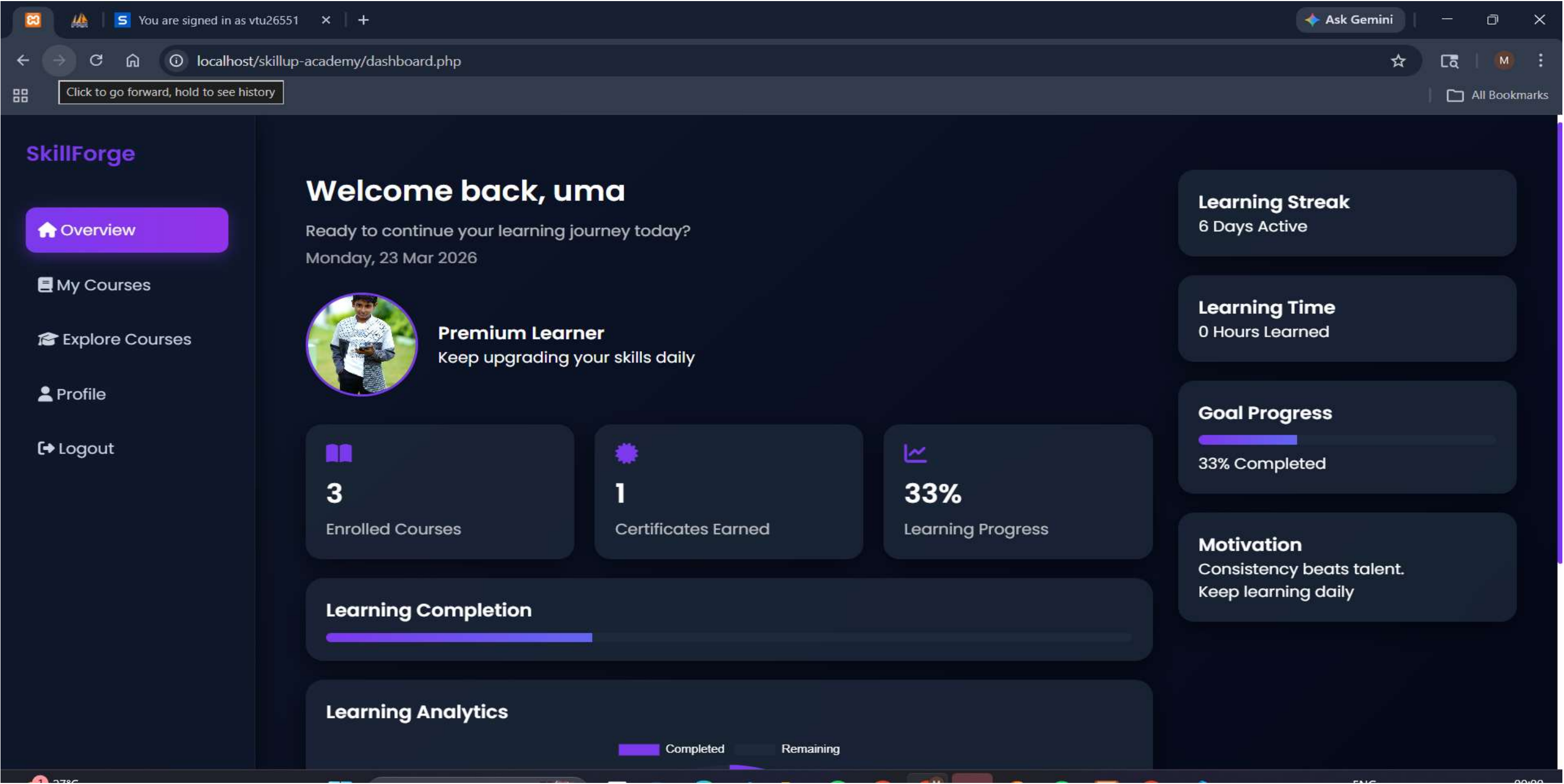
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ADVANTAGES & LIMITATIONS



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Advantages

- Easy access to courses anytime, anywhere
- Organized learning materials (videos & notes)
- Automatic progress tracking
- Simple course management for admin
- Quiz and certificate support

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Limitations

- Requires internet connection
- Limited offline access
- Basic security features
- Not optimized for very large users
- Needs more advanced features in future

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Thank you